Ps 05

Erik An

<2026-01-18 Sun>

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1 Problem 13 DONE

1.1 Task

Let

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix} \in \mathbb{R}^{3\times 3}, \quad b = \begin{bmatrix} 3 \\ 7 \\ 10 \end{bmatrix} \in \mathbb{R}^3.$$

(a) Perform the first step of Gaussian elimination with partial pivoting in the first column, i.e., look for an appropriate row j in your working column that contains a non-zero element, and then swap the first and j th row. Write down the permutation matrix P_1 representing the row swap that is performed (if any), and the resulting matrix $P_1 A$.

(b) Continue the elimination with partial pivoting until you obtain matrices P, L, U such that

$$PA = LU$$

, where P is a permutation matrix, L is unit lower triangular, and U is upper triangular. Write down P, L, and U explicitly.

(c) Solve $Ax = b$ using the factorization $PA = LU$ by solving $Ly = Pb$, $Ux = y$. Present y and r.

1.2 Solution

- Part (a)

  Given matrix A:

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix}$$

  Step 1: Partial Pivoting

  We examine the first column to find the pivot element with the largest absolute value.

$$|a_{11}| = 0, \quad |a_{21}| = 2, \quad |a_{31}| = 4$$

  Since the maximum value is in the third row, we must swap row 1 and row 3 ($R_1 \leftrightarrow R_3$). The permutation matrix P_1 corresponding to this swap is obtained by applying the same operation to the identity matrix:

$$P_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

  We now compute the resulting matrix $P_1 A$:

$$
\begin{aligned}
P_1 A &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix} \\
&= \begin{bmatrix} 4 & 1 & 8 \\ 2 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}
\end{aligned}
$$

Thus, the first step of Gaussian elimination with partial pivoting yields:

$$
P_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad P_1 A = \begin{bmatrix} 4 & 1 & 8 \\ 2 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}
$$

- Part (b)
  - Step 0: Given

$$
A = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix}
$$

  - Step 1: Pivot

$$
\begin{aligned}
P_1 A &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix} \qquad \Big(\text{Swap } R_1, R_3\Big) \\
&= \begin{bmatrix} 4 & 1 & 8 \\ 2 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}
\end{aligned}
$$

  - Step 2: Eliminate column 1

$$
= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{L_1} \underbrace{\begin{bmatrix} 4 & 1 & 8 \\ 0 & \frac{3}{2} & -1 \\ 0 & 2 & 1 \end{bmatrix}}_{U_1} \qquad \Big(R_2 \leftarrow R_2 - \frac{1}{2} R_1\Big)
$$

  - Step 3: Pivot col 2 (Swap R2, R3)

$$
\begin{aligned}
P_{total}A &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} A \qquad \Big(\text{Swap } R_2, R_3 \text{ in } U_1 \text{ and } L_1\Big) \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & \frac{3}{2} & -1 \end{bmatrix}
\end{aligned}
$$

- Step 4: Eliminate column 2

$$
= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{3}{4} & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{7}{4} \end{bmatrix}}_{U}
$$

- Step 5: Conclusion

$$
\begin{aligned}
P &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\
L &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{3}{4} & 1 \end{bmatrix} \\
U &= \begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{7}{4} \end{bmatrix}
\end{aligned}
$$

- Part (c)

  Given:

$$
Ax = b
$$

$$
\begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 10 \end{bmatrix}
$$

- **Forward Substitution** ($Ly = Pb$)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{3}{4} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{3}{4} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 7 \end{bmatrix}$$

* (i) From the first row:

$$y_1 = 10$$

* (ii) From the second row:

$$y_2 = 3$$

* (iii) From the third row:

$$\frac{1}{2}y_1 + \frac{3}{4}y_2 + y_3 = 7 \implies y_3 = 7 - 5 - \frac{9}{4} \implies y_3 = -\frac{1}{4}$$

Final Solution

$$y = \begin{bmatrix} 10 \\ 3 \\ -\frac{1}{4} \end{bmatrix}$$

- **Backward Substitution** ($Ux = y$)

$$\begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{7}{4} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ -\frac{1}{4} \end{bmatrix}$$

* (i) From the third row:

$$-\frac{7}{4}x_3 = -\frac{1}{4} \implies x_3 = \frac{1}{7}$$

* (ii) From the second row:

$$2x_2 + x_3 = 3 \implies 2x_2 = \frac{20}{14} \implies x_2 = \frac{10}{7}$$

* (iii) From the first row:

$$4x_1 + x_2 + 8x_3 = 10 \implies 4x_1 = 10 - \frac{10}{7} - \frac{8}{7} \implies 4x_1 = \frac{52}{7} \implies x_1 = \frac{13}{7}$$

Final Solution

$$x = \begin{bmatrix} \frac{13}{7} \\ \frac{10}{7} \\ \frac{1}{7} \end{bmatrix}$$

2 Problem 14 DONE

2.1 Task

(a) Let $\sigma = (3, 1, 2)$ and $\tau = (2, 1)$ be permutations of $\{1, 2, 3\}$. Write down the corresponding permutation matrices P_σ and P_τ acting on row vectors via left multiplication, i.e. $P_\sigma A$ permutes the rows of A by σ.

(b) Compute $P_\tau P_\sigma$ and identify the permutation ρ such that $P_\tau P_\sigma = P_\rho$. Explain briefly (one or two sentences) how this relates to composition of permutations.

(c) Prove that for every permutation matrix P one has $P^{-1} = P^T$.

2.2 Solution

- Part (a)

$$\sigma = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$
$$\tau = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Part (b)

  I compute the matrix product $P_\tau P_\sigma$:

$$\begin{aligned} P_\tau P_\sigma &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

  The resulting matrix has 1s at positions $(1, 1)$, $(2, 3)$, $(3, 2)$. This corresponds to the permutation:

$$\rho = (1, 3, 2)$$

  Explanation: The multiplication of permutation matrices corresponds to the composition of permutations. The matrix product $P_\tau P_\sigma$ represents applying the permutation σ first, followed by the permutation τ, which results in the combined permutation ρ.

- Part (c)

  Let $P \in \mathbb{R}^{n\times n}$ be a permutation matrix of size $n \times n$.

  Hence,

  $(P^\top P)_{ij} = (\text{row i of } P^\top) \cdot (\text{column j of } P)$, where row i of $P^\top$ is a column i of P.

  Therefore, $P^\top P$ is dotting column i of P with column j of P. Because P is a permutation matrix, each row and column has exactly one 1 and the rest are 0s.

  Hence, the dot product of a column with itself results in a 1 ($i = j$), while dot product a column with the others results in a 0 ($i \neq j$), leading to $P^\top P = I \implies P^\top = P^{-1}$ by definition of P^{-1}.

3 Problem 15 DONE

3.1 Task

Let

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{bmatrix} \in \mathbb{R}^{3\times 3}, \quad b^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad b^{(2)} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

(a) Compute an LU factorization with pivoting, i.e. find P, L, U such that $PA = LU$.

(b) Using the same P, L, U, solve $Ax = b^{(1)}$ and $Ax = b^{(2)}$. Show the two solutions.

3.2 Solution

- Part (a)

  - Step 0: Given

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{bmatrix}$$

  - Step 1: Pivot

$$\begin{aligned} P_1 A &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 3 & 4 \\ 3 & 1 & 1 \\ 3 & 1 & 5 \end{bmatrix} \end{aligned}$$

- Step 2: Eliminate column 1

$$
\begin{aligned}
P_1 A &= \begin{bmatrix} 6 & 3 & 4 \\ 3 & 1 & 1 \\ 3 & 1 & 5 \end{bmatrix} \\
&= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{L_1} \underbrace{\begin{bmatrix} 6 & 3 & 4 \\ 0 & -\frac{1}{2} & -1 \\ 3 & 1 & 5 \end{bmatrix}}_{U_{\text{partial}}} \qquad \left(R_2 \leftarrow R_2 - \frac{1}{2} R_1\right) \\
&= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix}}_{L_{\text{partial}}} \underbrace{\begin{bmatrix} 6 & 3 & 4 \\ 0 & -\frac{1}{2} & -1 \\ 0 & -\frac{1}{2} & 3 \end{bmatrix}}_{U_{\text{partial}}} \qquad \left(R_3 \leftarrow R_3 - \frac{1}{2} R_1\right)
\end{aligned}
$$

- Step 3: Eliminate column 2

$$
PA = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 6 & 3 & 4 \\ 0 & -\frac{1}{2} & -1 \\ 0 & 0 & 4 \end{bmatrix}}_{U} \qquad (R_3 \leftarrow R_3 - 1 \cdot R_2)
$$

- Step 4: Conclusion

$$
\begin{aligned}
P &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
L &= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 1 & 1 \end{bmatrix} \\
U &= \begin{bmatrix} 6 & 3 & 4 \\ 0 & -\frac{1}{2} & -1 \\ 0 & 0 & 4 \end{bmatrix}
\end{aligned}
$$

- Part (b)

  I should solve $Ax = b$ using the factorisation $PA = LU$ in two stages:

  1. Solve $Ly = Pb$ for y (Forward Substitution)
  2. Solve $Ux = y$ for x (Backward Substitution)

3. Case 1: $b^{(1)} = [1, 2, 3]^\top$

$$Pb^{(1)} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

- Step 1: Forward Substitution

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

  * (i) From the first row:

$$y_1 = 2$$

  * (ii) From the second row:

$$\frac{1}{2}y_1 + y_2 = 1 \implies y_2 = 0$$

  * (iii) From the third row:

$$\frac{1}{2}y_1 + y_2 + y_3 = 3 \implies y_3 = 2$$

  Final Solution:

$$y = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

- Step 2: Backward Substitution

$$\begin{bmatrix} 6 & 3 & 4 \\ 0 & -\frac{1}{2} & -1 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

  * (i) From the third row:

$$4x_3 = 2 \implies x_3 = \frac{1}{2}$$

  * (ii) From the second row:

$$-\frac{1}{2}x_2 - x_3 = 0 \implies x_2 = -1$$

  * (iii) From the first row:

$$6x_1 + 3x_2 + 4x_3 = 2 \implies 6x_1 = 2 + 3 - 2 \implies x_1 = \frac{1}{2}$$

Final Solution:

$$x = \begin{bmatrix} \frac{1}{2} \\ -1 \\ \frac{1}{2} \end{bmatrix}$$

4. Case 2: $b^{(2)} = [2, 0, 1]^\top$

$$Pb^{(2)} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

   - Step 1: Forward Substitution

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

     * (i) From the first row:

$$y_1 = 0$$

     * (ii) From the second row:

$$\frac{1}{2}y_1 + y_2 = 2 \implies y_2 = 2$$

     * (iii) From the third row:

$$\frac{1}{2}y_1 + y_2 + y_3 = 1 \implies y_3 = -1$$

     Final Solution:

$$y = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

   - Step 2: Backward Substitution

$$\begin{bmatrix} 6 & 3 & 4 \\ 0 & -\frac{1}{2} & -1 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

     * (i) From the third row:

$$4x_3 = -1 \implies x_3 = -\frac{1}{4}$$

     * (ii) From the second row:

$$-\frac{1}{2}x_2 - x_3 = 2 \implies -\frac{1}{2}x_2 = \frac{7}{4} \implies x_2 = -\frac{7}{2}$$

* (iii) From the first row:

$$6x_1 + 3x_2 + 4x_3 = 0 \implies 6x_1 = \frac{21}{2} + 1 \implies x_1 = \frac{23}{12}$$

Final Solution:

$$x = \begin{bmatrix} \frac{23}{12} \\ -\frac{7}{2} \\ -\frac{1}{4} \end{bmatrix}$$

4 Problem 16 DONE

4.1 Task

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

(a) Perform Gaussian elimination with partial pivoting and determine the rank of A from the resulting echelon form.

(b) For the two right-hand sides

$$b^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad b^{(2)} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

decide for each system $Ax = b^{(j)}$ whether it is consistent, i.e., whether it has a solution. If consistent, describe the full solution set.

4.2 Solution

- Part (a)
  - Step 0: Given

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

  - Step 1: Pivoting

$$
\begin{aligned}
P_1 A &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}
\end{aligned}
$$

- Step 2: Eliminate column 1

$$
\begin{aligned}
P_1 A &= \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \qquad \left(R_2 \leftarrow R_2 - \frac{1}{2} R_1 \right) \\
&= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \qquad \left(R_3 \leftarrow R_3 - \frac{1}{2} R_1 \right)
\end{aligned}
$$

- Step 3: Pivoting
  We swap row 2 and row 3. This applies to the rows of U and the multipliers in column 1 of L.

$$
PA = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \mathbf{\frac{1}{2}} & 1 & 0 \\ \mathbf{\frac{1}{2}} & 0 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 4 & 6 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}}_{U}
$$

- Step 4: Conclusion

$$
\begin{aligned}
P &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \\
L &= \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{\frac{1}{2}} & 1 & 0 \\ \mathbf{\frac{1}{2}} & 0 & 1 \end{bmatrix} \\
U &= \begin{bmatrix} 2 & 4 & 6 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}
\end{aligned}
$$

The matrix U has 2 non-zero rows. Therefore, $\mathbf{Rank(A) = 2}$.

- Part (b)

  Since the matrix is singular (Rank < 3) leads to the following cases:

  1. No solution
  2. Infinite solutions
  3. Case 1: $b^{(1)} = [1, 2, 1]^\top$
     - Step 1: Forward Substitution

$$Ly = Pb^{(1)}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

       * (i) from the first row:

$$y_1 = 2$$

       * (ii) from the second row:

$$\frac{1}{2}y_1 + y_2 = 1 \implies y_2 = 0$$

       * (iii) from the third row:

$$\frac{1}{2}y_1 + y_3 = 1 \implies y_3 = 0$$

     Final Solution:

$$y = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

     - Step 2: Backward Substitution

$$\begin{bmatrix} 2 & 4 & 6 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

       * (i) From the third row:

$$0 = 0. \text{ I set } x_3 = t : t \in \mathbb{R}$$

       * (ii) From the second row:

$$-x_2 - 2x_3 = 0 \implies x_2 = -2t$$

* (iii) From the first row:

$$2x_1 + 4x_2 + 6x_3 = 2 \implies 2x_1 = 2 + 2t \implies x_1 = 1 + t$$

Final Solution Set:

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

4. Case 2: $b^{(2)} = [1, 3, 1]^\top$
   - Step 1: Forward Substitution

$$Ly = Pb^{(2)}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

   * (i) from the first row:

$$y_1 = 3$$

   * (ii) from the second row:

$$\frac{1}{2}y_1 + y_2 = 1 \implies y_2 = -\frac{1}{2}$$

   * (iii) from the third row:

$$\frac{1}{2}y_1 + y_3 = 1 \implies y_3 = -\frac{1}{2}$$

   Final Solution:

$$y = \begin{bmatrix} 3 \\ -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

   - Step 2: Backward Substitution

$$\begin{bmatrix} 2 & 4 & 6 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

   * (i) From the third row:

$$0 = -\frac{1}{2}$$

   Conclusion:
   Since $0 \neq -\frac{1}{2}$, the system is inconsistent, and thus has no solutions.

5 Problem 17 DONE

5.1 Task

Let $\mathbb{F} \in \{\mathbb{Q}, \mathbb{R}, \mathbb{C}\}$ and assume that in step k of Gaussian elimination with partial pivoting the pivot row is chosen such that

$$|a_{kk}^{(k)}| = max_{i \in \{k,...,n\}} |a_{ik}^{(k)}|$$

where $a_{ik}^{(k)}$ denotes the entries of the matrix after the first $k-1$ elimination steps.

(a) Show that the elimination multipliers

$$l_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}, \quad i = k+1, ...n,$$

satisfy $|l_{ik}| \leq 1$.

(b) Conclude that in the resulting factorization $PA = LU$ obtained by column pivoting, all subdiagonal entries of L satisfy $|L_{ik}| \leq 1$ for $i > k$.

5.2 Solution

- Part (a)

  Let $a_{ik}^{(k)}$ denote the entries of column i, below row k. According to the *partial pivoting rule*, I must choose a pivot row p (where $pgek$) such that $a_{ik}^{(k)}$ is maximal. I then swap row p and row k.

  Hence, now, the new diagonal element satisfies:

$$\forall i \in \{k+1, ..., n\}, |a_{kk}^{(k)}| \geq |a_{ik}^{(k)}|$$

  The elimination multiplier l_{ik} for the row i is defined as:

$$l_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$$

  Taking the absolute value:

$$|l_{ik}| = \left|\frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}\right| = \frac{|a_{ik}^{(k)}|}{|a_{kk}^{(k)}|}$$

  Since the numetator is less or equal to denominator, it follows that:

$$|l_{ik}| \leq 1$$

- Part (b)

  The matrix L is unit lower triangular. Its strict lower triangular part consists fully of multipliers l_{ik} calculated during the elimination step.

  Although the subsequent pivoting steps may permute the rows of the matrix, this corresponds to permuting rows of the multipliers already stored in L. Specifically, if the rows are swapped, this results in the change of entries positions inside of the matrix, but their magnitudes remain unchanged.

  Since we established in Part (a) that every multiplier generated is bounded by $|l_{ik}| \leq 1$ at the moment of its creation, and since row permutations only move these values around without altering their magnitude, every subdiagonal entry L_{ik} in the final matrix satisfies:

$$|L_{ik}| \leq 1$$

6 Problem 18 DONE

6.1 Task

A sensor produces one measurement per hour over an 8-hour cycle, yielding the data

$$y = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \\ 2 \\ 4 \\ 8 \\ 6 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^8$$

A common "data science" model for periodic effects is to fit a small number of Fourier features. Here we use the constant term and the harmonic of frequency 2:

$$\hat{y}_t = a_0 + a_c cos(\frac{\pi t}{2}) + a_s sin(\frac{\pi t}{2}), \quad t = 0, 1, ..., 7,$$

where a_0, a_c $a_s \in \mathbb{R}$ are to be chosen such that the squared error

$$\sum_{t=0}^{7} (\hat{y}_t - y_t)^2$$

is minimal.

Define the design matrix $X \in \mathbb{R}^{8\times 3}$ by

$$X_{t+1,1} = 1, \quad X_{t+1,2} = cos(\frac{\pi t}{2}), \quad X_{t+1,3} = sin(\frac{\pi t}{2}), \quad t = 0, ..., 7$$

so that $\hat{y} = Xa$ with $a = (a_0, a_c, a_s)^T$.

(a) Write down X explicitly by using the values

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$$cos(\frac{\pi t}{2}) \in \{-1, 0, 1\}, \quad sin(\frac{\pi t}{2}) \in \{-1, 0, 1\}$$

(b) Compute $X^T X$ and show that it is diagonal. (Hint: You only need the sums $\sum_{t=0}^{7} cos(\pi t/2)$, $\sum_{t=0}^{7} sin(\pi t/2)$, $\sum_{t=0}^{7} cos(\pi t/2) sin(\pi t/2)$, $\sum_{t=0}^{7} cos^2(\pi t/2)$, $\sum_{t=0}^{7} sin^2(\pi t/2)$.

(c) Derive the (so called) normal equations

$$(X^T X)a = X^T y$$

and solve them for a_0, a_c, u_s.

(d) Compute the fitted signal $\hat{y} = Xa$ and the residual $r = y - \hat{y}$. Give the squared residual norm $||r||_2^2 = \sum_{t=0}^{7} r_t^2$.

6.2 Solution

- Part (a)

  I have 8 observations. The model is:

  $$\hat{y}_i = a_0 \cdot (1) + a_c \cdot \cos(\frac{\pi t}{2}) + a_s \cdot \sin(\frac{\pi t}{2}) \qquad t = 0, 1, ..., 7.$$

  This corresponds to the matrix multiplication $\hat{y} = Xa$, where the columns of X are the values attached to a_0, a_c, a_s respecitvely.

  - Column 1: always 1
  - Column 2: $\cos(\frac{\pi t}{2})$
    - $t = 0 \rightarrow \cos(0) = 1$
    - $t = 1 \rightarrow \cos(\frac{\pi}{2}) = 0$
    - $t = 2 \rightarrow \cos(\pi) = -1$
    - $t = 3 \rightarrow \cos(\frac{3\pi}{2}) = 0$
    - The pattern 1, 0, -1, 0 repeats.
  - Column 3: $\sin(\frac{\pi t}{2})$
    - $t = 0 \rightarrow \sin(0) = 0$
    - $t = 1 \rightarrow \sin(\frac{\pi}{2}) = 1$
    - $t = 2 \rightarrow \sin(\pi) = 0$
    - $t = 3 \rightarrow \sin(\frac{3\pi}{2}) = -1$
    - The pattern 0, 1, 0, -1 repeats.

  The Matrix X:

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

- Part (b)

  The matrix product $X^T X$ calculated the dot product between the columns of X, and results in a 3×3 matrix.

  - Diagonal Elements
    * Position (1,1): Sum of squared 1s

      $$\sum_{t=0}^{8} 1^2 = 8$$

    * Position (2,2): Sum of squared cosines

      $$(1^2 + 0^2 + (-1)^2 + 0^2 + 1^2 + 0^2 + (-1)^2 + 0^2) = 4$$

    * Position (3,3): Sum of squared sines

      $$(0^2 + 1^2 + 0^2 + (-1)^2 + 0^2 + 1^2 + 0^2 + (-1)^2) = 4$$

  - Non-Diagonal Elements
    * Col 1 $\cdot$ Col 2: Sum of cosines $(1 + 0 - 1 + 0 \ldots)$. The sum is 0.
    * Col 1 $\cdot$ Col 3: Sum of sines $(0 + 1 + 0 - 1 \ldots)$. The sum is 0.
    * Col 2 $\cdot$ Col 3: Sum of $\cos(t) \cdot \sin(t)$. Looking at the rows of X, whenever cosine is non-zero (1 or -1), sine is 0. Whenever sine is non-zero, cosine is 0. Their product is always 0. Sum is 0.

  Result:

  $$X^T X = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

- Part (c)

  To find the optimal coefficients a, we use the geometric interpretation of least squares. We seek the vector $\hat{y} = Xa$ in the column space of X that is closest to y. The error vector (residual) $r = y - Xa$ must be orthogonal to the column space of X.

  Mathematically, this means X^T multiplied by the residual is zero:

  $$X^T(y - Xa) = 0$$

Distributing the transpose matrix:

$$X^T y - X^T X a = 0$$

Rearranging terms gives the **Normal Equations**:

$$(X^T X)a = X^T y$$

We already have $(X^T X)$ from Part (b). Now we calculate the vector $X^T y$. This involves taking the dot product of each column of X with the data vector $y = [8, 7, 2, 4, 8, 6, 2, 3]^T$.

- First component (Sum of y):

$$1 \cdot (8 + 7 + 2 + 4 + 8 + 6 + 2 + 3) = 40$$

- Second component $(X_{\text{col2}} \cdot y)$:

$$1(8) + 0(7) + (-1)(2) + 0(4) + 1(8) + 0(6) + (-1)(2) + 0(3)$$

$$= 8 - 2 + 8 - 2 = 12$$

- Third component $(X_{\text{col3}} \cdot y)$:

$$0(8) + 1(7) + 0(2) + (-1)(4) + 0(8) + 1(6) + 0(2) + (-1)(3)$$

$$= 7 - 4 + 6 - 3 = 6$$

So, $X^T y = \begin{bmatrix} 40 \\ 12 \\ 6 \end{bmatrix}$.

Now we solve the system $(X^T X)a = X^T y$:

$$\begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_c \\ a_s \end{bmatrix} = \begin{bmatrix} 40 \\ 12 \\ 6 \end{bmatrix}$$

Because the matrix is diagonal, we can solve each row independently:

- $8a_0 = 40 \implies a_0 = 5$
- $4a_c = 12 \implies a_c = 3$
- $4a_s = 6 \implies a_s = 1.5$

Solution:

$$a = \begin{bmatrix} 5 \\ 3 \\ 1.5 \end{bmatrix}$$

- Part (d)

- The Fitted Signal ($\hat{y} = Xa$): We plug our coefficients back into the model equation: $\hat{y}_t = 5 + 3\cos(\frac{\pi t}{2}) + 1.5\sin(\frac{\pi t}{2})$.
  * $t = 0 : 5 + 3(1) + 1.5(0) = \mathbf{8}$
  * $t = 1 : 5 + 3(0) + 1.5(1) = \mathbf{6.5}$
  * $t = 2 : 5 + 3(-1) + 1.5(0) = \mathbf{2}$
  * $t = 3 : 5 + 3(0) + 1.5(-1) = \mathbf{3.5}$
  * $t = 4 : 5 + 3(1) + 1.5(0) = \mathbf{8}$
  * $t = 5 : 5 + 3(0) + 1.5(1) = \mathbf{6.5}$
  * $t = 6 : 5 + 3(-1) + 1.5(0) = \mathbf{2}$
  * $t = 7 : 5 + 3(0) + 1.5(-1) = \mathbf{3.5}$

$$\hat{y} = \begin{bmatrix} 8 \\ 6.5 \\ 2 \\ 3.5 \\ 8 \\ 6.5 \\ 2 \\ 3.5 \end{bmatrix}$$

- The Residual ($r = y - \hat{y}$): Subtract the fitted values from the observed values ($y = [8, 7, 2, 4, 8, 6, 2, 3]^T$).
  * $8 - 8 = 0$
  * $7 - 6.5 = 0.5$
  * $2 - 2 = 0$
  * $4 - 3.5 = 0.5$
  * $8 - 8 = 0$
  * $6 - 6.5 = -0.5$
  * $2 - 2 = 0$
  * $3 - 3.5 = -0.5$

$$r = \begin{bmatrix} 0 \\ 0.5 \\ 0 \\ 0.5 \\ 0 \\ -0.5 \\ 0 \\ -0.5 \end{bmatrix}$$

- Squared Residual Norm ($||r||_2^2$): Sum the squares of the residual vector.

$$||r||_2^2 = 0^2 + (0.5)^2 + 0^2 + (0.5)^2 + 0^2 + (-0.5)^2 + 0^2 + (-0.5)^2$$
$$||r||_2^2 = 0 + 0.25 + 0 + 0.25 + 0 + 0.25 + 0 + 0.25$$
$$||r||_2^2 = 1$$